



**D22 AIM COMPRESSION LIMITER  
PRESSING MACHINE.**  
**Operator / Maintenance Manual**

**Manufactured by:**  
Paveway Automation Pvt. Ltd.  
Sanghvi Steel Complex,  
128/2, TELCO Road,  
Chinchwad,  
Pune - 411019



## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

Gat No 148/150, Chakan,Pune-410501  
(Near Dwarka School ,Village- Ambethan ,Taluka- Khed)



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## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

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### SYSTEM INFORMATION:

Name:

D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

Customer:

Gat No 148/150, Chakan,Pune-410501(Near Dwarka School ,Village -Ambethan ,Taluka -Khed)

Manufacturing Date:

November, 2021.

Supply Voltage:

230 VAC, 50/60 Hz

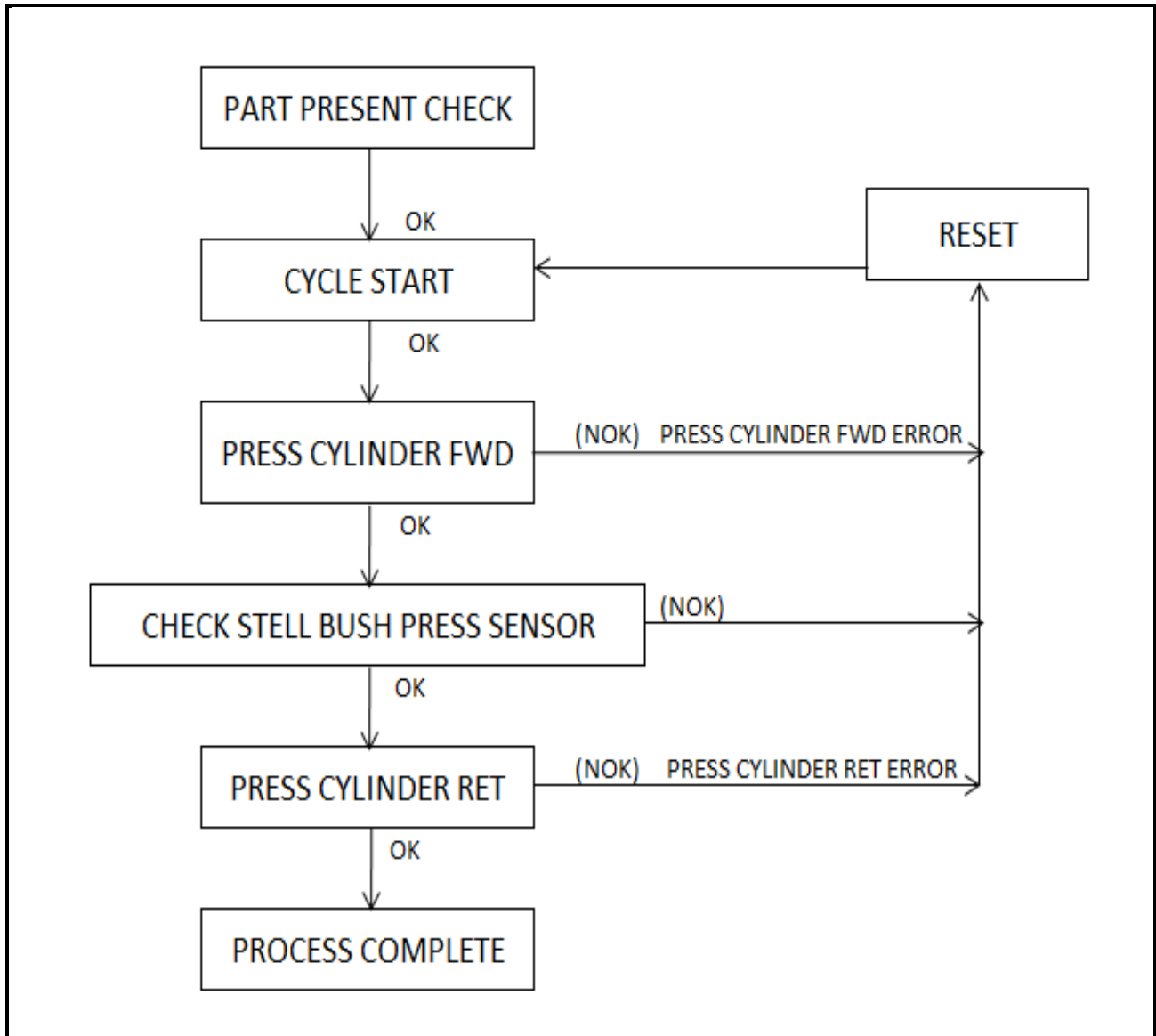
Manufacturer:

**Paveway Automation Pvt. Ltd.**  
Sanghvi Steel Complex,  
128/2, Telco Road, Chinchwad,  
Pune- 411019.  
tel: 020-27450840

Email ID :

[applications@pavewayautomation.com](mailto:applications@pavewayautomation.com)

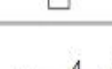
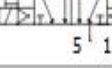
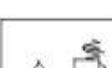
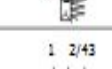
**PROCESS SEQUENCE:**



## WIRING DIAGRAMS:

### SYMBOL DESCRIPTION

### SYMBOL DESCRIPTION

| Symbol                                                                              | Symbol description    | Symbol                                                                              | Symbol description           |
|-------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------|------------------------------|
|    | Earthing              |    | Buzzer                       |
|    | 2 Pole MCB            |   | Double acting solenoid valve |
|    | 2 Pole on/off switch  |  | Double acting solenoid valve |
|   | Lamp                  |  | Double acting solenoid valve |
|  | Tubelight             |  | Double acting solenoid valve |
|  | Fan                   |  | Double acting solenoid valve |
|  | SMPS                  |  | Double acting solenoid valve |
|  | 1 pole MCB            |  | Double acting solenoid valve |
|  | Push button           |  | Double acting solenoid valve |
|  | Relay coil            |  | Double acting solenoid valve |
|  | Emergency stop button |  | Double acting solenoid valve |
|  | Proximity sensor      |  | Double acting solenoid valve |
|  | Airoff                |  | Double acting solenoid valve |
|                                                                                     |                       |  | Digital pressure switch      |





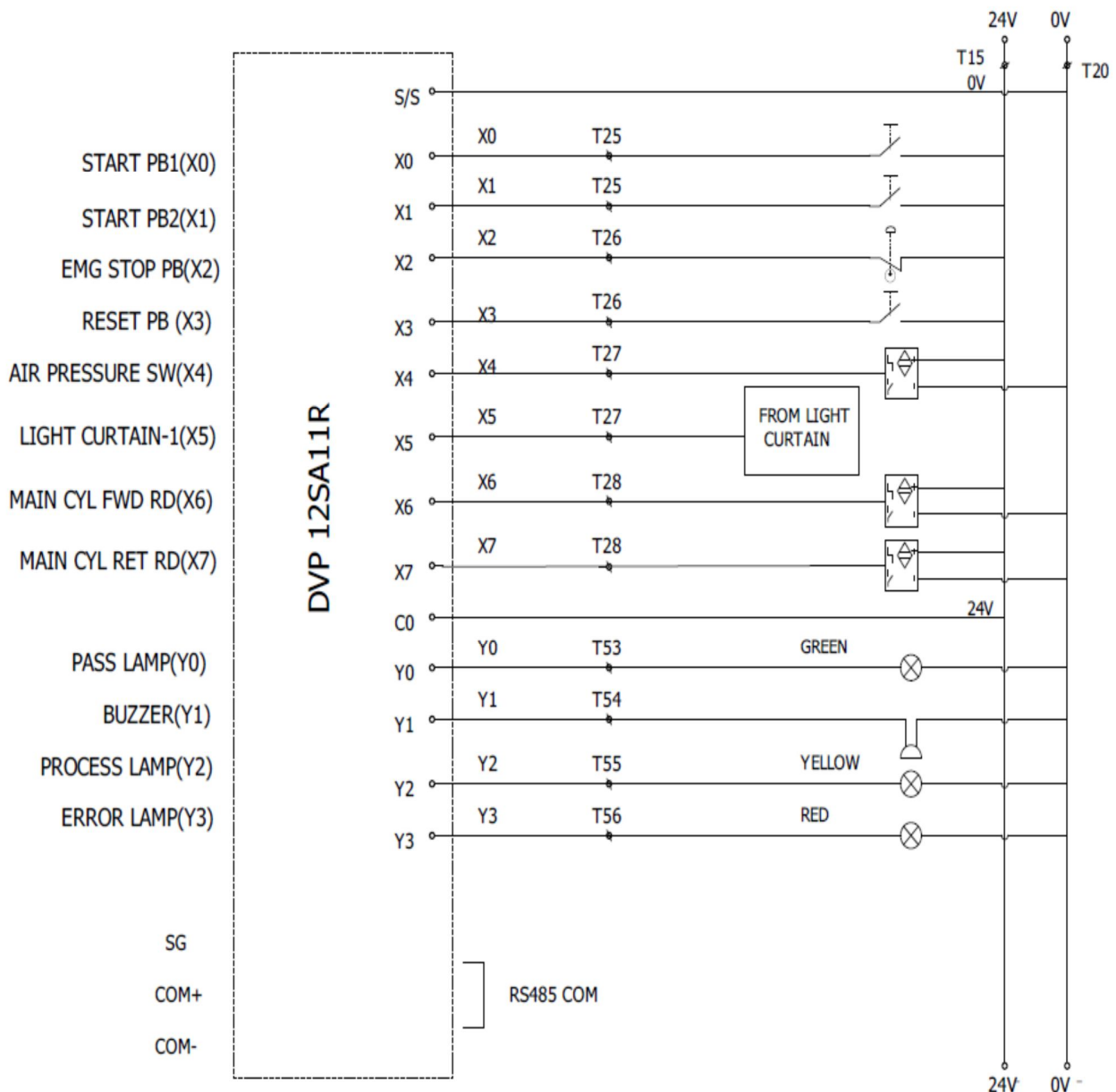
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Gat No 148/150, Chakan,Pune-410501  
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### MAIN PLC MODULE-DVP 12SA11R

#### PLC MAIN MODULE: DVP 12SA11R





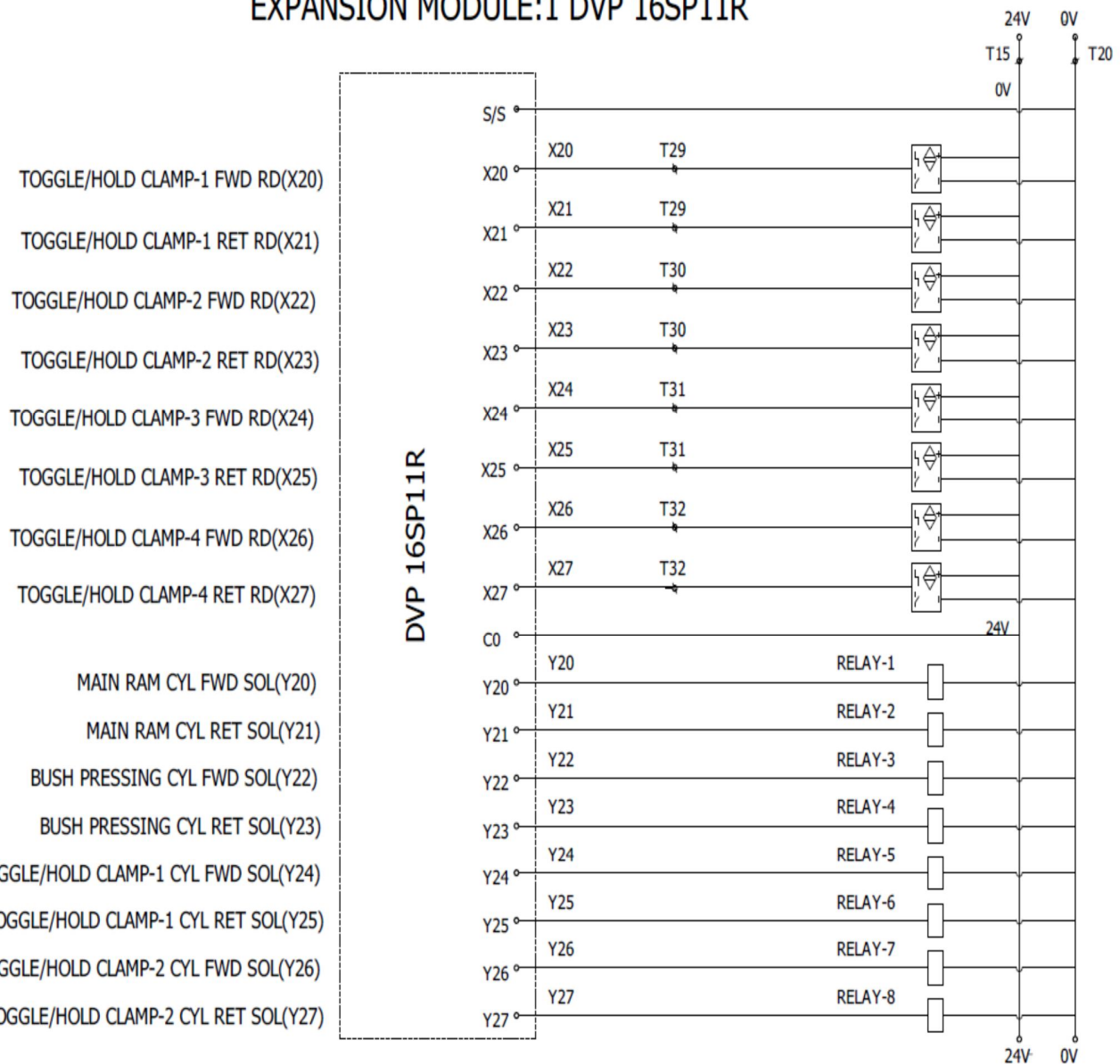
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### EXPANSION MODULE-1 DVP16SP11R

#### EXPANSION MODULE:1 DVP 16SP11R





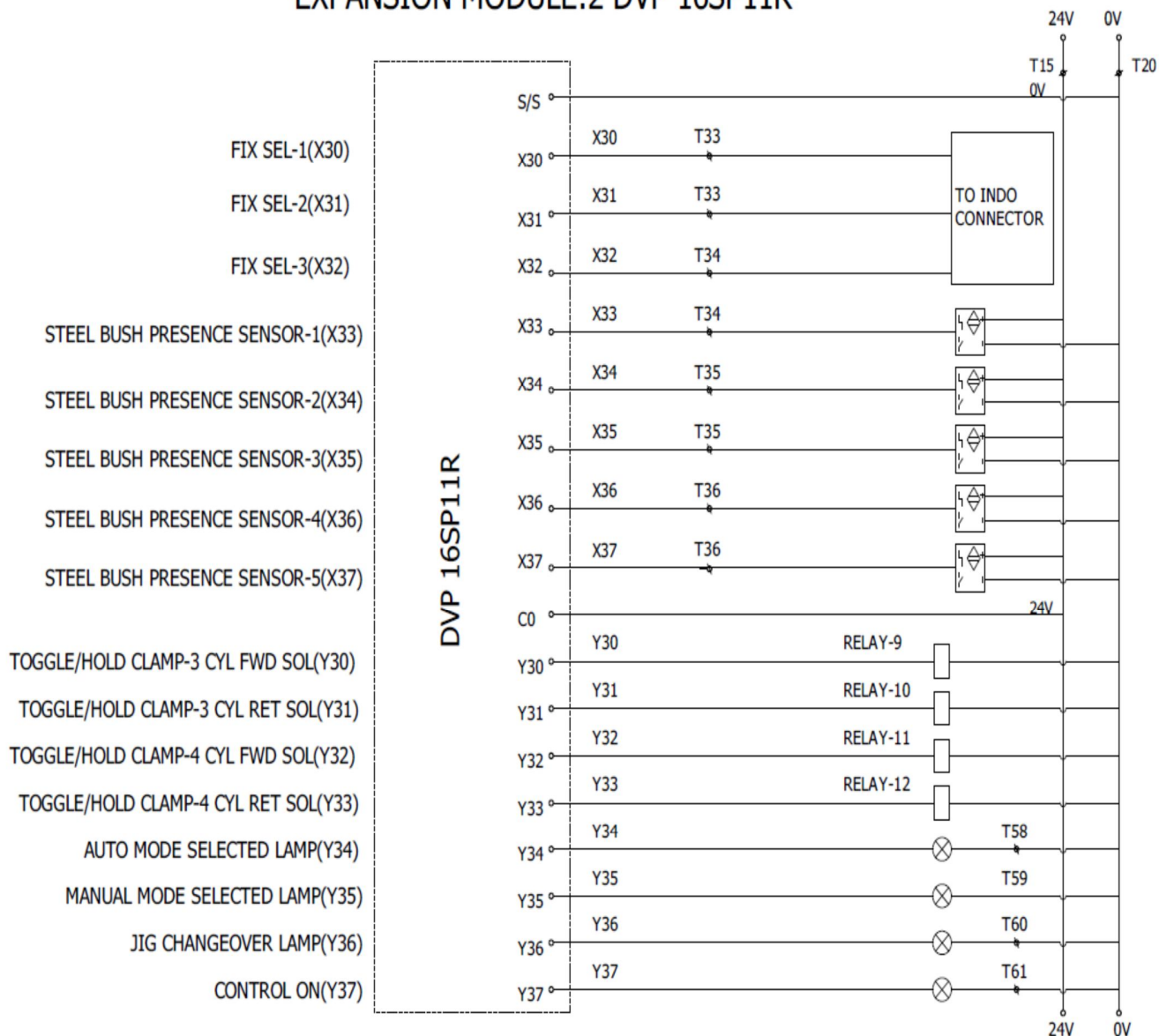
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### EXPANSION MODULE-2 DVP 16SP11R

#### EXPANSION MODULE:2 DVP 16SP11R





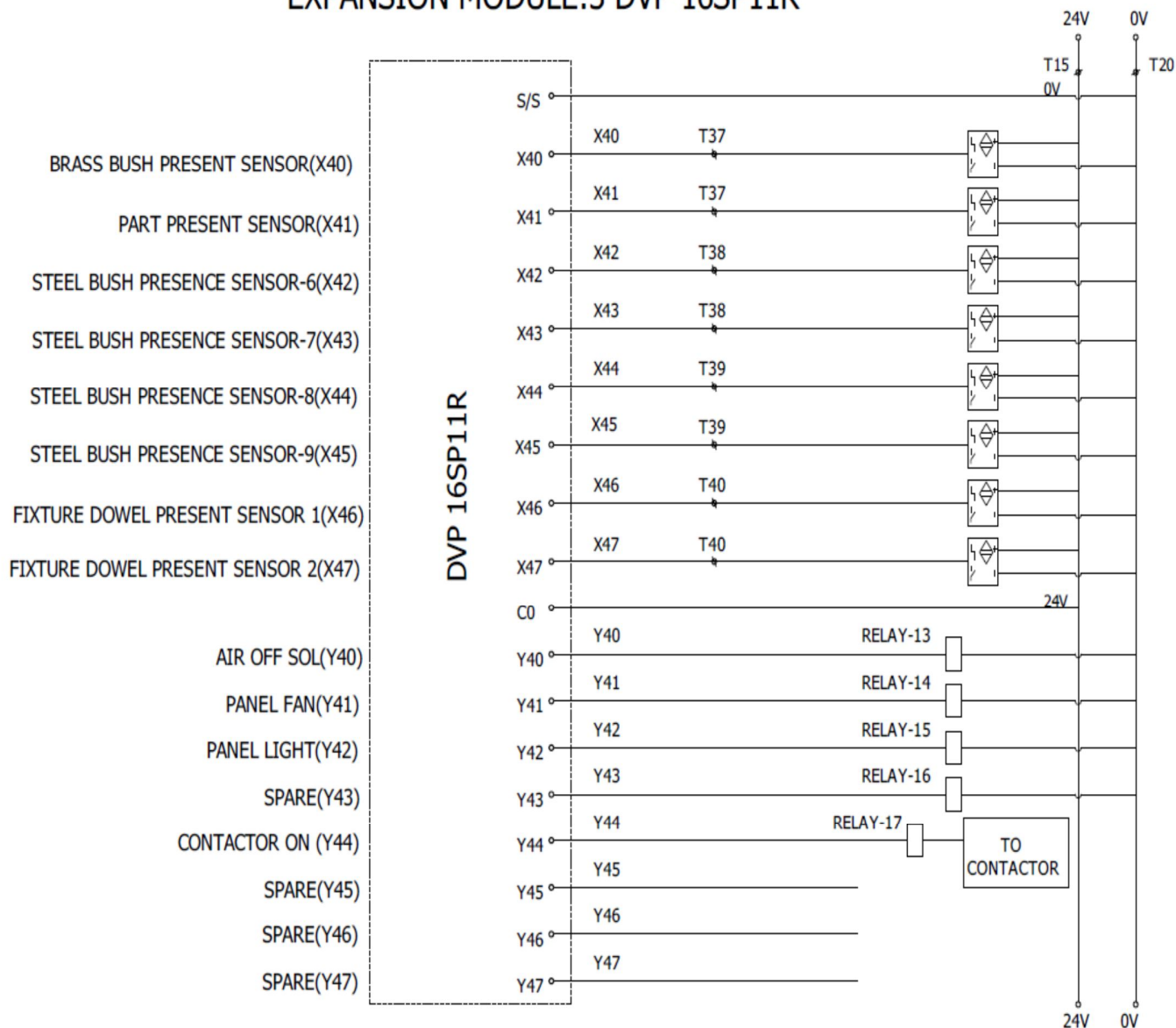
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### EXPANSION MODULE-3 DVP 16SP11R

#### EXPANSION MODULE:3 DVP 16SP11R





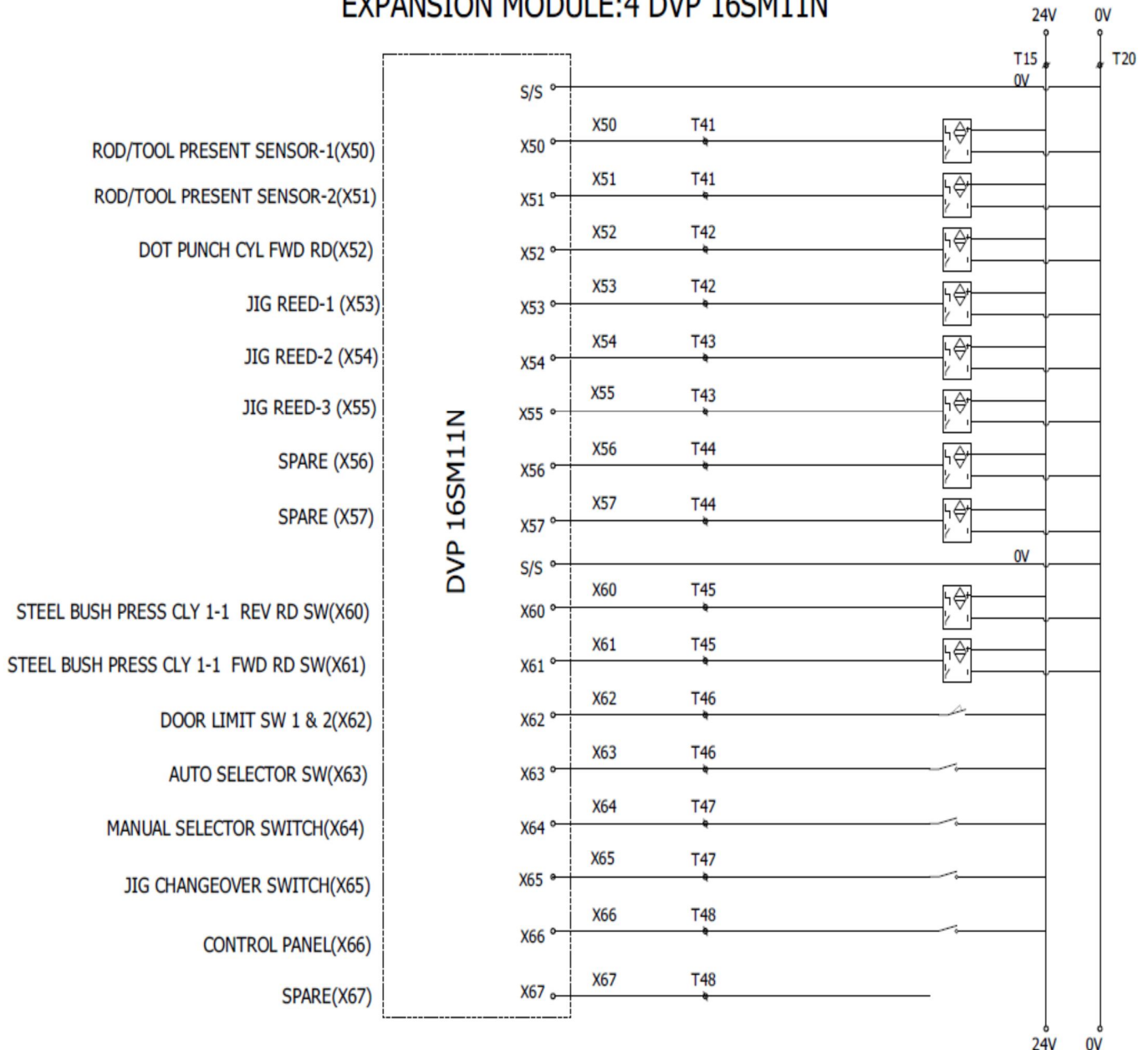
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### EXPANSION MODULE: 4 DVP 16SM11N

#### EXPANSION MODULE:4 DVP 16SM11N





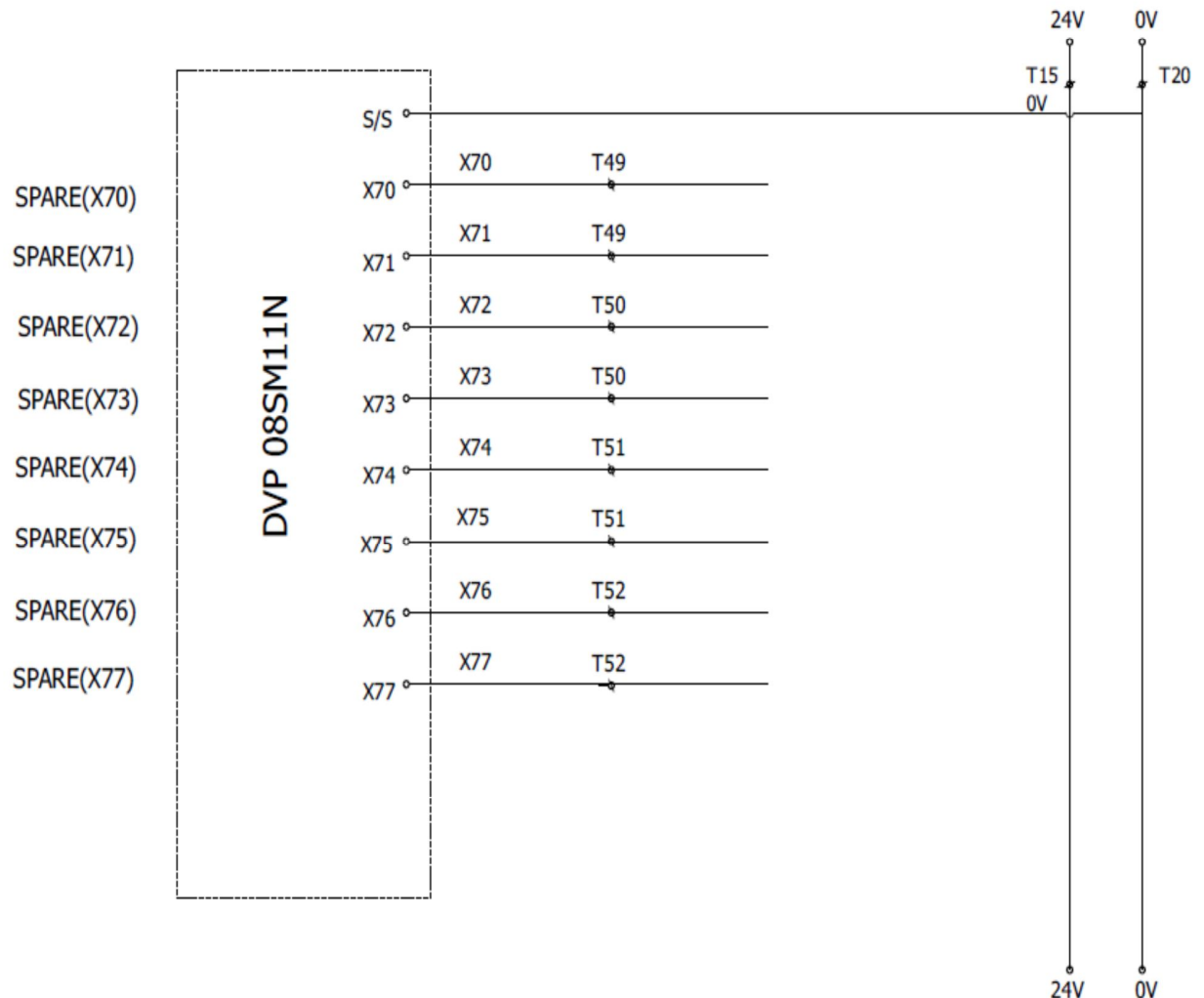
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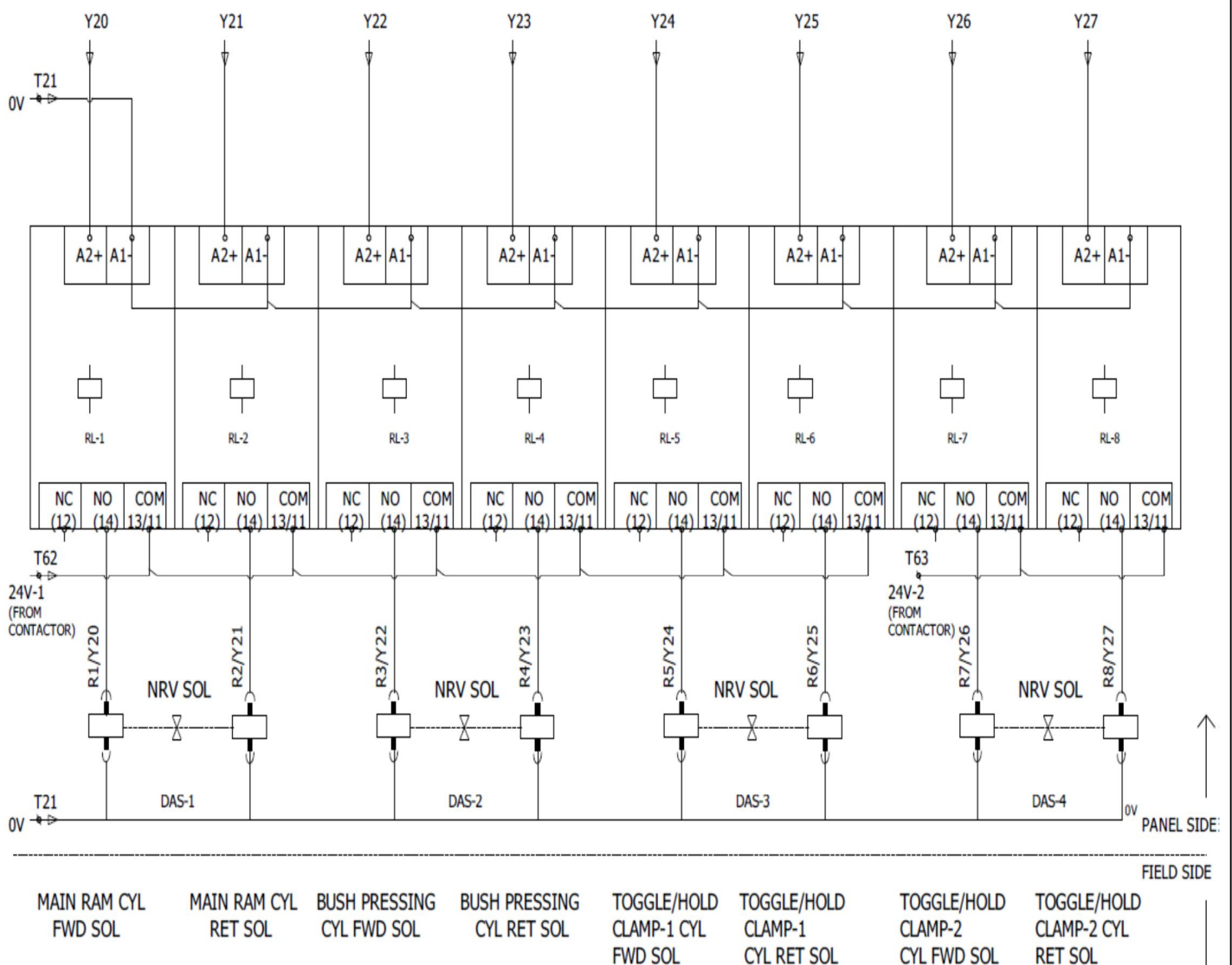
### EXPANSION MODULE: 5 DVP 08SM11N

#### EXPANSION MODULE:5 DVP 08SM11N



## RELAY DIAGRAM-1

### RELAY DIAGRAM-1





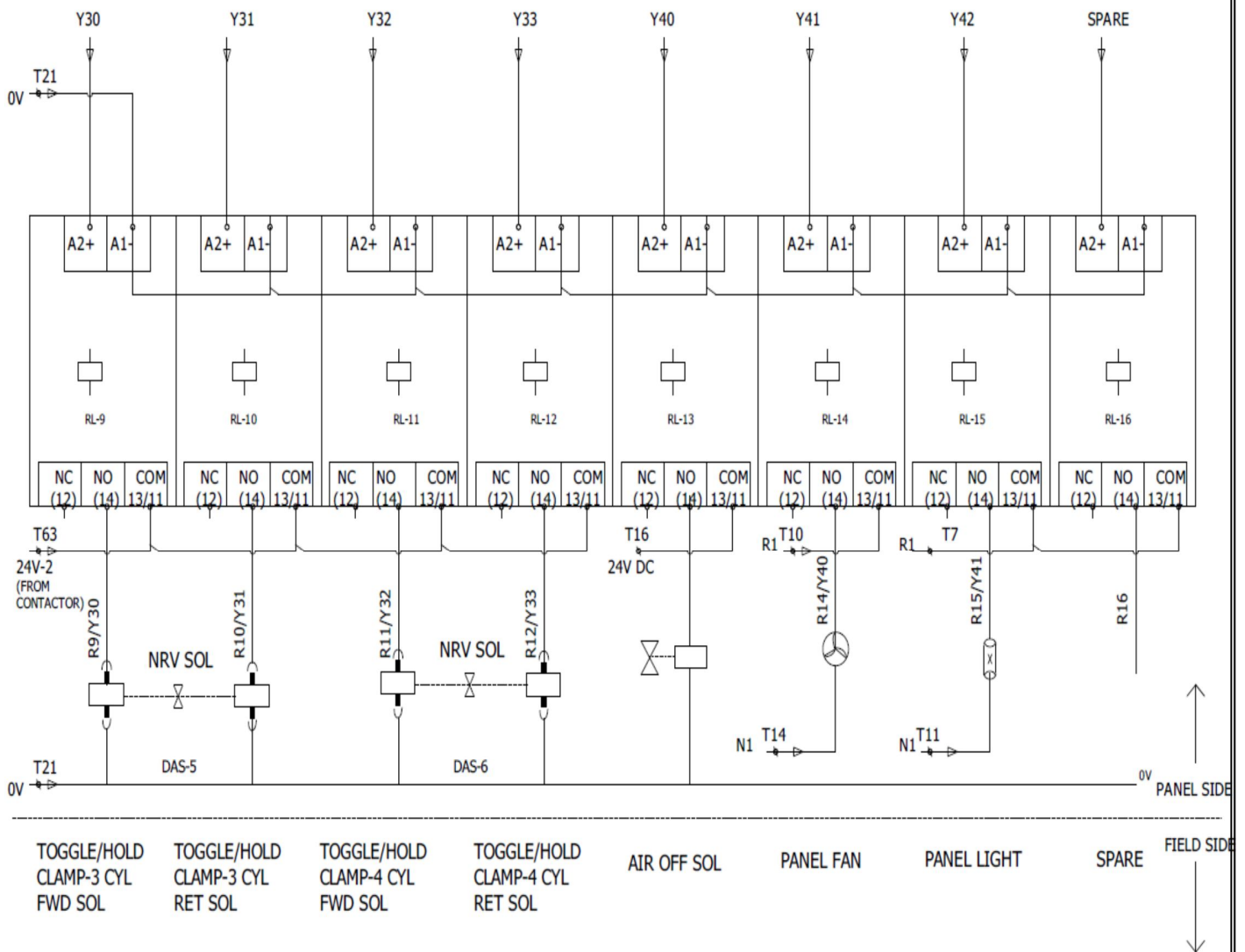
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### RELAY DIAGRAM-2

### RELAY DIAGRAM-2







## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

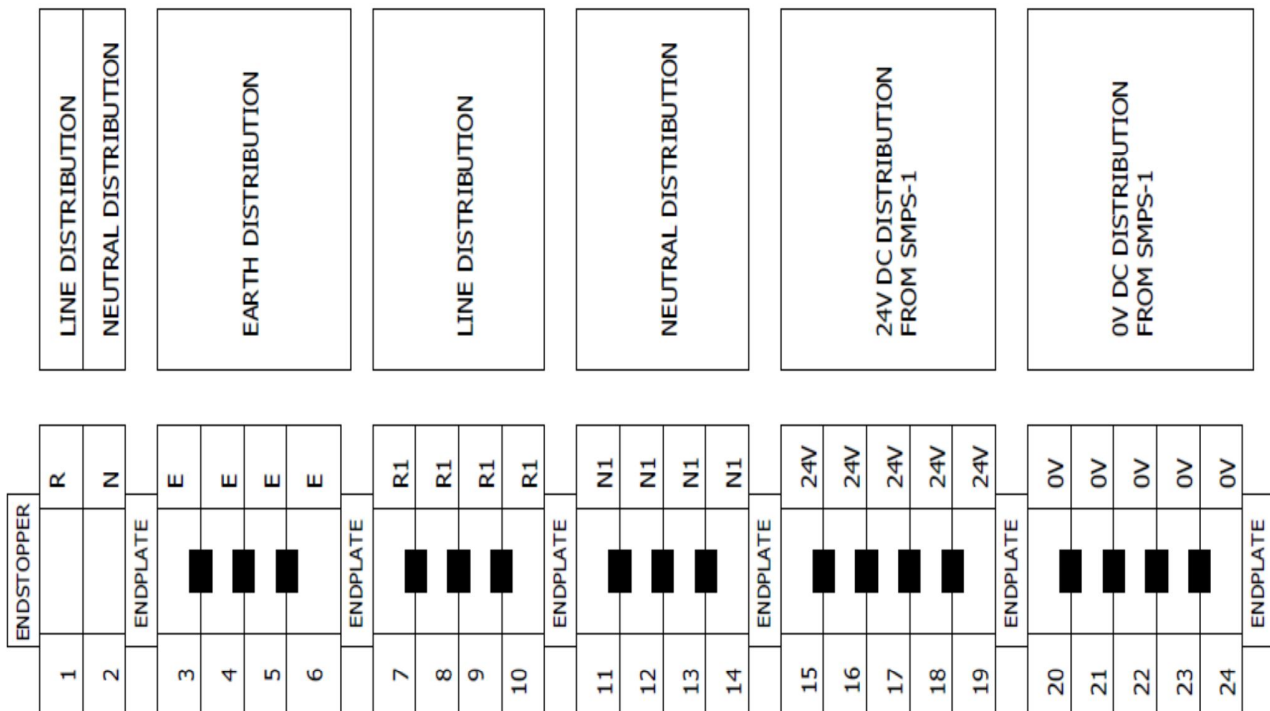
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### PANEL CONNECTOR DETAILS-1

#### PANEL CONNECTOR DETAILS-1

FIELD SIDE



PANEL SIDE

## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

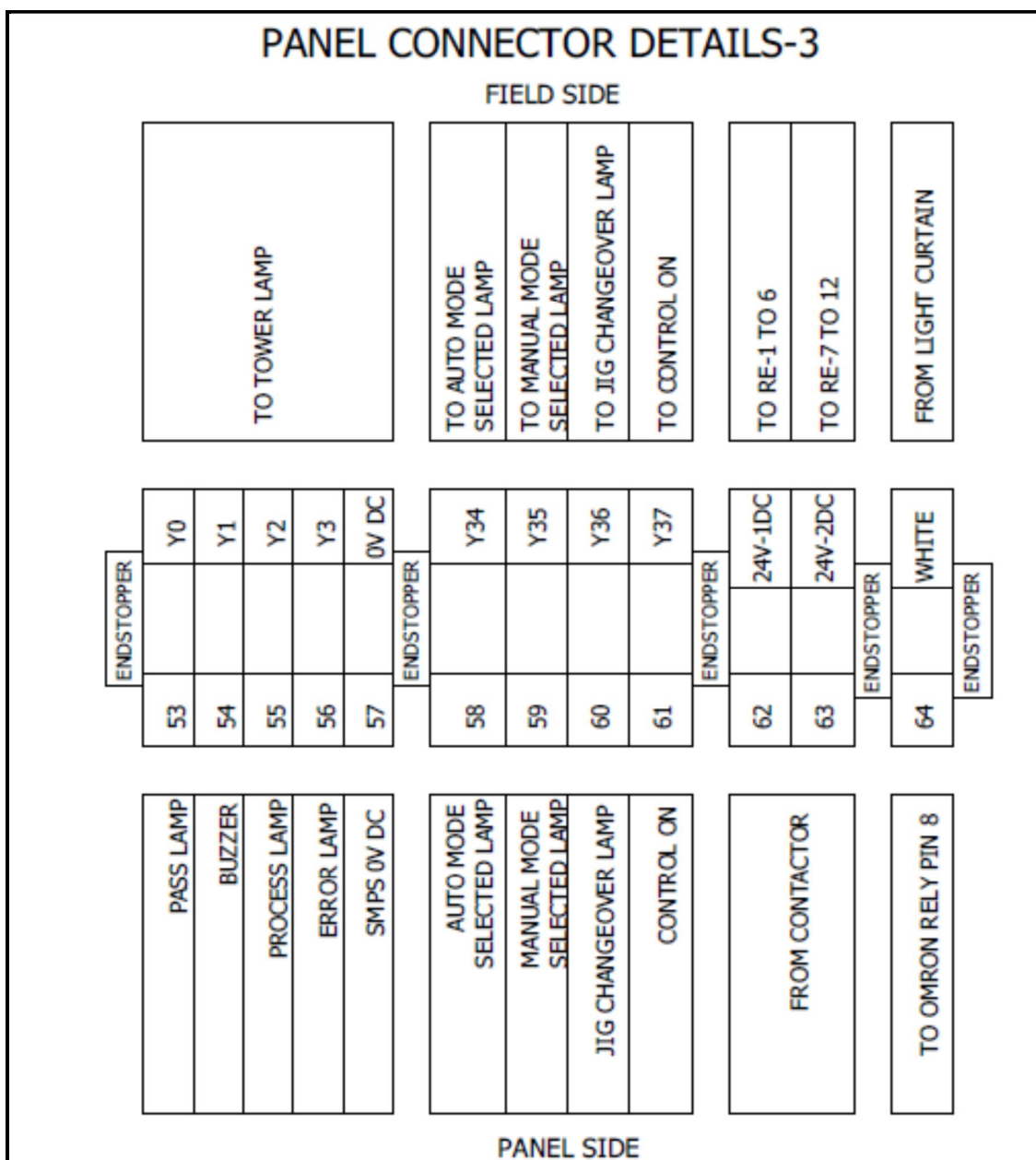
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### PANEL CONNECTOR DETAILS-2

#### PANEL CONNECTOR DETAILS-2

| PANEL SIDE                              |     |     |    | FIELD SIDE |     |                                         |  |
|-----------------------------------------|-----|-----|----|------------|-----|-----------------------------------------|--|
|                                         |     |     |    | ENDSTOPPER |     |                                         |  |
| START PB2(X1)                           | X1  | X0  | 25 | X0         | X1  | START PB1(X0)                           |  |
| RESET PB (X3)                           | X3  | X2  | 26 | X2         | X3  | EMG STOP PB(X2)                         |  |
| LIGHT CURTAIN-1(X5)                     | X5  | X4  | 27 | X4         | X5  | AIR PRESSURE SW(X4)                     |  |
| MAIN CYL RET RD(X7)                     | X7  | X6  | 28 | X6         | X7  | MAIN CYL FWD RD(X6)                     |  |
| TOGGLE/HOLD CLAMP-1 RET RD(X21)         | X21 | X20 | 29 | X20        | X21 | TOGGLE/HOLD CLAMP-1 FWD RD(X20)         |  |
| TOGGLE/HOLD CLAMP-2 RET RD(X23)         | X23 | X22 | 30 | X22        | X23 | TOGGLE/HOLD CLAMP-2 FWD RD(X22)         |  |
| TOGGLE/HOLD CLAMP-3 RET RD(X25)         | X25 | X24 | 31 | X24        | X25 | TOGGLE/HOLD CLAMP-3 FWD RD(X24)         |  |
| TOGGLE/HOLD CLAMP-4 RET RD(X27)         | X27 | X26 | 32 | X26        | X27 | TOGGLE/HOLD CLAMP-4 FWD RD(X26)         |  |
| FIX SEL-2(X31)                          | X31 | X30 | 33 | X30        | X31 | FIX SEL-1(X30)                          |  |
| STEEL BUSH PRESENCE SENSOR-1(X33)       | X33 | X32 | 34 | X32        | X33 | FIX SEL-3(X32)                          |  |
| STEEL BUSH PRESENCE SENSOR-3(X35)       | X35 | X34 | 35 | X34        | X35 | STEEL BUSH PRESENCE SENSOR-2(X34)       |  |
| STEEL BUSH PRESENCE SENSOR-5(X37)       | X37 | X36 | 36 | X36        | X37 | STEEL BUSH PRESENCE SENSOR-4(X36)       |  |
|                                         |     |     |    | ENDSTOPPER |     |                                         |  |
| PART PRESENT SENSOR(X41)                | X41 | X40 | 37 | X40        | X41 | BRASS BUSH PRESENT SENSOR(X40)          |  |
| STEEL BUSH PRESENCE SENSOR-7(X43)       | X43 | X42 | 38 | X42        | X43 | STEEL BUSH PRESENCE SENSOR-6(X42)       |  |
| STEEL BUSH PRESENCE SENSOR-9(X45)       | X45 | X44 | 39 | X44        | X45 | STEEL BUSH PRESENCE SENSOR-8(X44)       |  |
| FIXTURE DOWEL PRESENT SENSOR 2(X47)     | X47 | X46 | 40 | X46        | X47 | FIXTURE DOWEL PRESENT SENSOR 1(X46)     |  |
| ROD/TOOL PRESENT SENSOR-2(X51)          | X51 | X50 | 41 | X50        | X51 | ROD/TOOL PRESENT SENSOR-1(X50)          |  |
| JIG REED-1 (X53)                        | X53 | X52 | 42 | X52        | X53 | DOT PUNCH CYL FWD RD(X52)               |  |
| JIG REED-3 (X55)                        | X55 | X54 | 43 | X54        | X55 | JIG REED-2 (X54)                        |  |
| SPARE(X57)                              | X57 | X56 | 44 | X56        | X57 | SPARE(X56)                              |  |
| STEEL BUSH PRESS CLY 1-1 FWD RD SW(X61) | X61 | X60 | 45 | X60        | X61 | STEEL BUSH PRESS CLY 1-1 REV RD SW(X60) |  |
| AUTO SELECTOR SW(X63)                   | X63 | X62 | 46 | X62        | X63 | DOOR LIMIT SW 1& 2(X62)                 |  |
| JIG CHANGEOVER SWITCH(X65)              | X65 | X64 | 47 | X64        | X65 | MANUAL SELECTOR SWITCH(X64)             |  |
| SPARE(X67)                              | X67 | X66 | 48 | X66        | X67 | CONTROL PANEL(X66)                      |  |
|                                         |     |     |    | ENDSTOPPER |     |                                         |  |
| SPARE(X71)                              | X71 | X70 | 49 | X70        | X71 | SPARE(X70)                              |  |
| SPARE(X73)                              | X73 | X72 | 50 | X72        | X73 | SPARE(X72)                              |  |
| SPARE(X75)                              | X75 | X74 | 51 | X74        | X75 | SPARE(X74)                              |  |
| SPARE(X77)                              | X77 | X76 | 52 | X76        | X77 | SPARE(X76)                              |  |

### PANEL CONNECTOR DETAILS-3





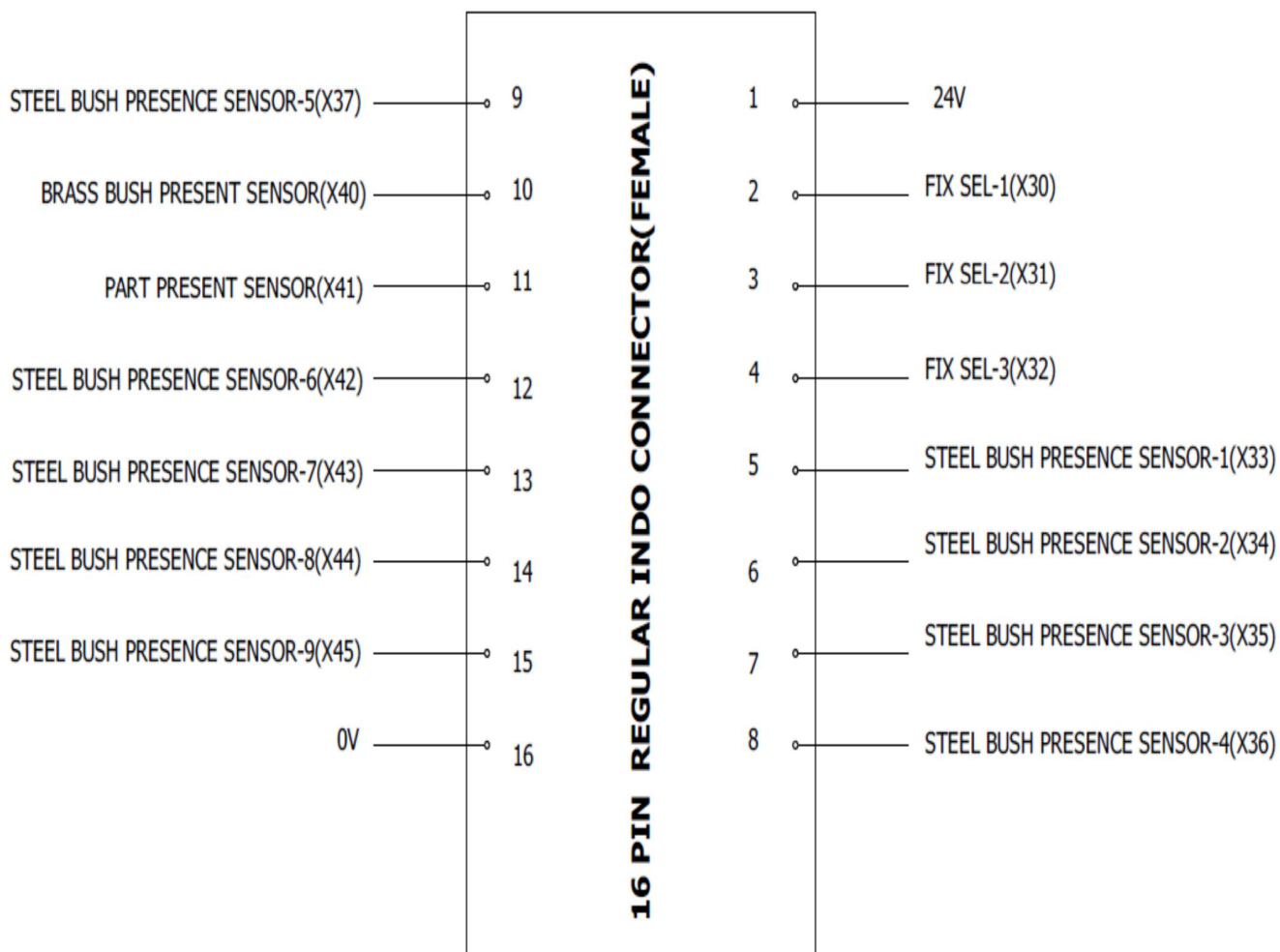
## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

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(Near Dwarka School ,Village- Ambethan ,Taluka-Khed)



### MACHINE SIDE INDO (FEMALE)

#### MACHINE SIDE 16 PIN INDO CONNECTOR(FEMALE)



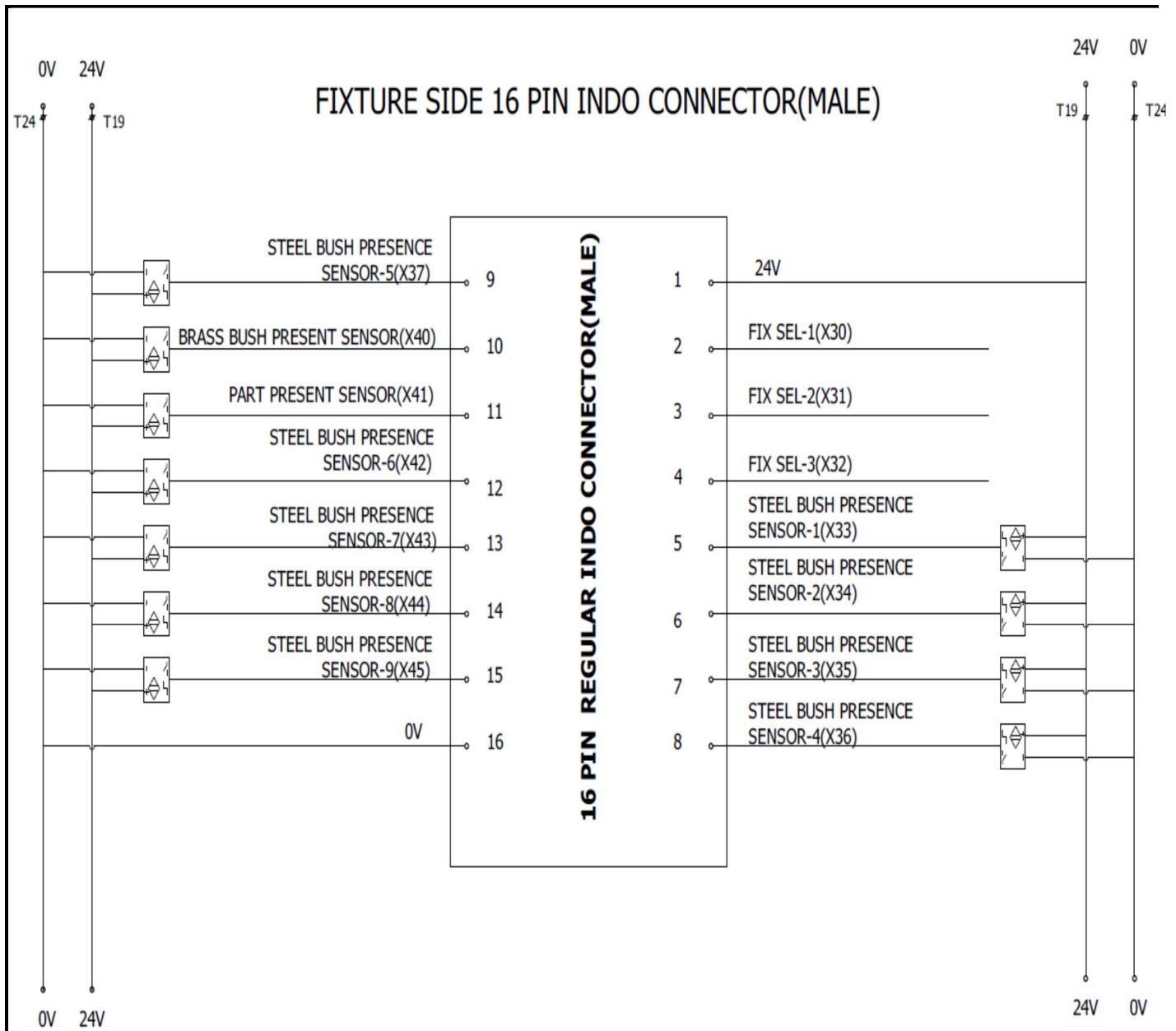


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### FIXTURE SIDE INDO (MALE)

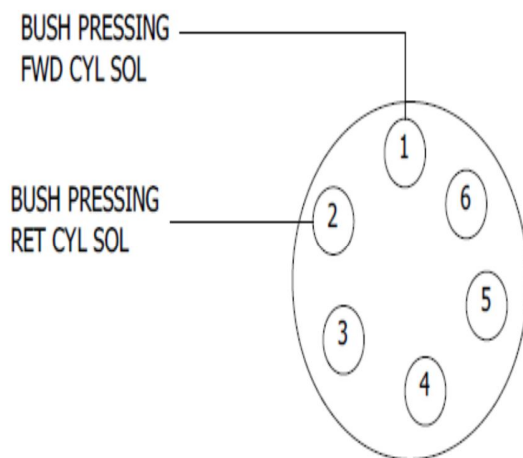




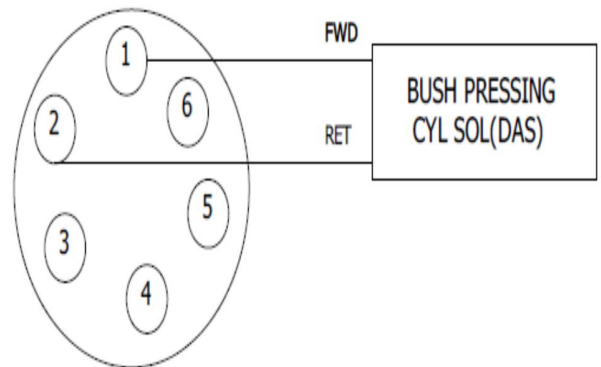
**PNEUMATIC CONNECTOR**

**PNEUMATIC CONNECTOR**

**PNEUMATIC CONNECTOR  
(MALE)**



**PNEUMATIC CONNECTOR  
(FEMALE)**





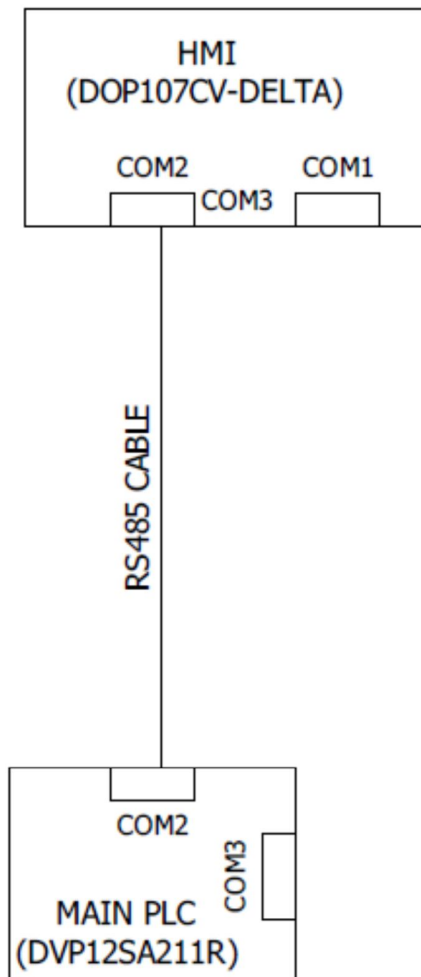
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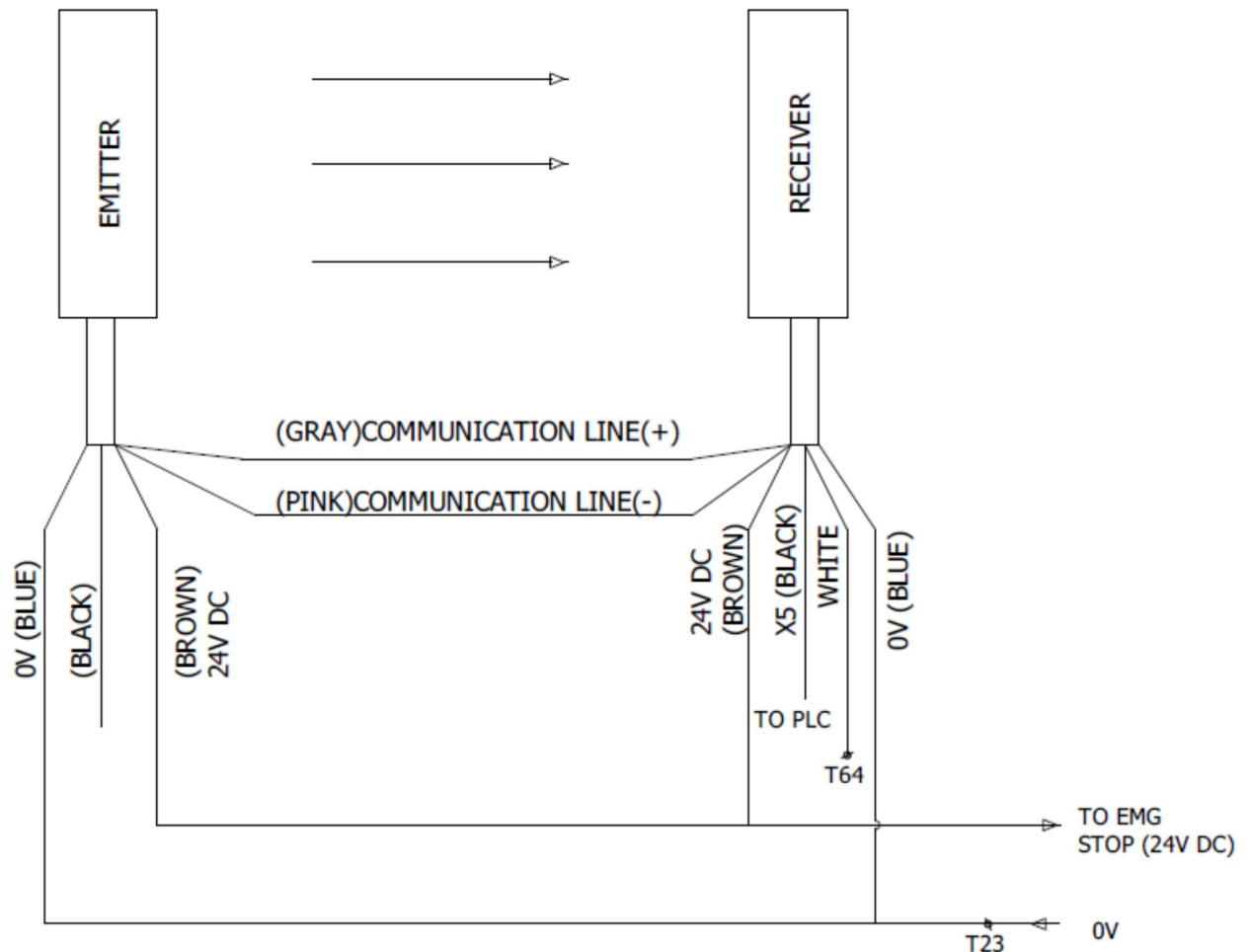
### COMMUNICATION ARCHITECTURE

## COMMUNICATION ARCHITECTURE



## LIGHT CURTAIN CONNECTIONS

### LIGHT CURTAIN CONNECTIONS (OMRON-F3SJ-E0385P25)





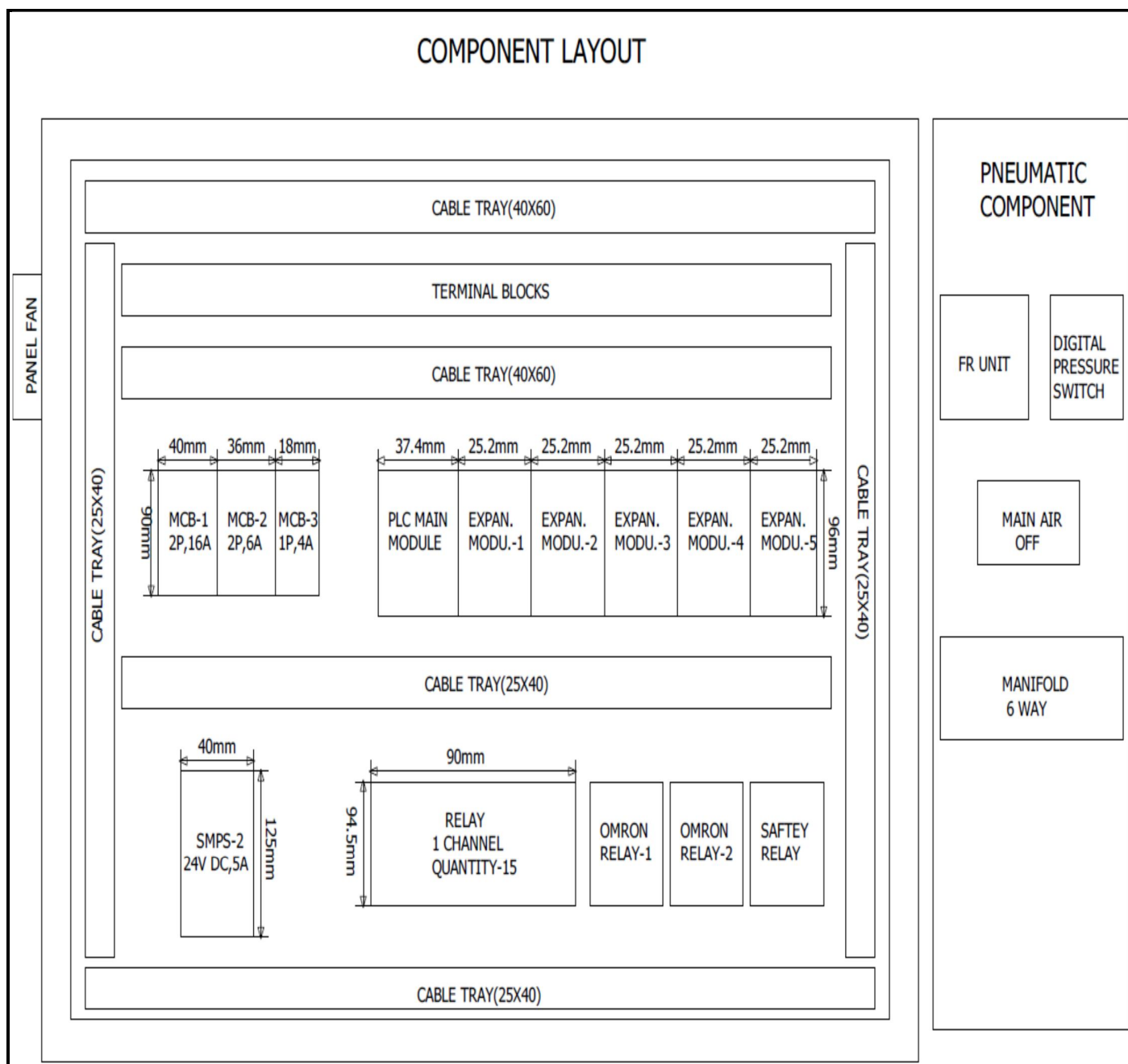
## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

Gat No 148/150, Chakan,Pune-410501  
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### COMPONENT LAYOUT

#### COMPONENT LAYOUT





## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

Gat No 148/150, Chakan,Pune-410501  
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### CABLE DETAILS

#### CABLE DETAILS

| CABLE            | CORE DIAMETER                                | COLOUR CODE  |               | MAKE     |
|------------------|----------------------------------------------|--------------|---------------|----------|
| POWER CABLE      | 1.5 SQ MM<br>3 CORE FLEXIBLE                 | RED          | LINE          | POLYCAB  |
|                  |                                              | BLACK        | NEUTRAL       |          |
|                  |                                              | YELLOW+GREEN | EARTH         |          |
| PANEL WIRING     | CONTROL WIRING<br>0.5 SQ. MM,1 CORE          | BLUE         | +24V DC       | RR CABLE |
|                  |                                              | WHITE        | 0V DC         |          |
|                  |                                              | GREY         | I/P OF PLC    |          |
|                  |                                              | ORANGE       | OUTPUT OF PLC |          |
|                  | POWER WIRING<br>1 SQ. MM,1 CORE              | RED          | LINE          |          |
|                  |                                              | BLACK        | NEUTRAL       |          |
|                  |                                              | YELLOW+GREEN | EARTH         |          |
| TOWER LAMP       | 0.25 SQ.MM<br>6 CORE,SHIELDED                | BLUE         | 0V DC         | STANDARD |
|                  |                                              | GRAY/WHITE   | SPARE         |          |
|                  |                                              | BROWN        | BUZZER        |          |
|                  |                                              | GREEN        | PASS LAMP     |          |
|                  |                                              | YELLOW       | PROCESS LAMP  |          |
|                  |                                              | PINK         | ERROR LAMP    |          |
| PLC TO HMI CABLE | RS 485 CABLE<br>2 CORE,0.25SQMM,<br>SHIELDED | BROWN        | D+            | STANDARD |
|                  |                                              | WHITE        | D-            |          |



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### **BILL OF MATERIAL:**

#### **ELECTRICAL PANEL MATERIAL**

| SN | COMPONENT NAME      | DESCRIPTION             | PART NO.         | MAKE            | QTY. |
|----|---------------------|-------------------------|------------------|-----------------|------|
| 1  | Control panel       |                         |                  | Eldon           | 1    |
| 2  | Plc                 | 8DI/4DO                 | DVP 12SA11R      | Delta           | 1    |
| 3  | Expansion module    | 8DI/8DO                 | DVP 16SP11R      | Delta           | 3    |
| 4  | Expansion module    | 16DI                    | DVP 16SM11N      | Delta           | 1    |
| 5  | Expansion module    | 8DI                     | DVP 08SM11N      | Delta           | 1    |
| 6  | Hmi                 | 7"                      | DOP107CV         | Delta           | 1    |
| 7  | Smps                | 24 VDC,5A               | S8VK-C12024      | OMRON           | 1    |
| 9  | On/off switch       | 2P,16A                  | LB116            | L & T           | 1    |
| 10 | MCB (Pannel)        | 2P,16A                  | 5SL62167RC       | Simens          | 1    |
| 11 | Mcb (Smps input)    | 2P,6A                   | 5SL62067RC       | Simens          | 1    |
| 12 | Mcb (24vdc)         | 1P,4A                   | 5SL61047RC       | Simens          | 1    |
| 13 | Tower lamp 3-tier   | Steady,24 VDC           | LSL-03-01-RAG-PB | Leuze           | 1    |
| 14 | Relay               | 24V,6A                  | SOCY1-24D        | Connectwell     | 17   |
| 15 | Safety Relay        | 24V DC,3A               | MSR127T          | Allen-Bradley   | 1    |
| 16 | Relay               | 24VDC                   | LY2N             | Omron           | 1    |
| 17 | Power Contactor     | 24VDC,7A                | 3RT2015-1BB41    | Simens          | 1    |
| 18 | Push Button (Green) | Green Flush Type        | 3PLBRL           | Teknic          | 2    |
| 19 | Reset Button(red)   | Red Flush Type          | 3PLBRL           | Teknic          | 1    |
| 20 | Emergency Stop      | Latched                 |                  | Teknic          | 1    |
| 21 | Electric indo       | 16 Pin Female connector |                  | Indo electrical | 1    |
| 22 | Electric indo       | 16 Pin male connector   |                  | Indo electrical | 1    |
| 23 | NO Element          | 24 VDC                  |                  | Teknic          | 4    |
| 24 | NC Element          | 24 VDC                  |                  | Teknic          | 1    |
| 25 | PB Box              | 3Way                    |                  | Teknic          | 2    |



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|    |                             |                    |               |        |   |
|----|-----------------------------|--------------------|---------------|--------|---|
| 26 | Indication Lamp (Green)     | 230 VAC            |               | Teknic | 1 |
| 28 | Auto/Manual Selector switch |                    |               |        | 1 |
| 29 | Panel fan                   | 230VAC,18W,2650RPM | 12A230HSAC    | HICOOL | 1 |
| 30 | LED station Tube Light      | 230VAC             | LP-201-15W    | Lord's | 1 |
| 31 | 3 pin top                   | 6A                 | 38626         | Anchor | 1 |
| 32 | Station fan                 | 230VAC,38W,2600RPM | 17A230HSAC    | HICOOL | 2 |
| 33 | Light curtain               | 24VDC              | F3SJ-E0385P25 | OMRON  | 1 |

### PNEUMATIC MATERIAL

| SN | ITEM                      | PART NO.         | DESCRIPTION           | MAKE | QTY |
|----|---------------------------|------------------|-----------------------|------|-----|
| 1  | Reed Switch               | -                | 24VDC,PNP             | -    | 13  |
| 2  | PNEUMATIC CONNECTOR ROUND | -                | 6 pin Round connector | -    | 1   |
| 3  | Solenoid Valve (NRV)      | SY7320-5LZ-02    | 5/3" VALVE(DAS)       | SMC  | 6   |
| 4  | Manifold (NRV)            |                  | 6-Way                 |      | 1   |
| 5  | Air Off Solenoid Valve    | VT307-5DZ1-02    | 3-Port                | SMC  | 1   |
| 6  | FR Unit                   | AW20-02BG1-B     | 1/4"                  | SMC  | 1   |
| 7  | Digital Pressure Switch   | ISE30AF-01-P-LA2 | 0-10 Bar              | SMC  | 1   |



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### I/O LIST:

| MODULES       | I/P ADDR | INPUTS                         | O/P ADDR | OUTPUTS                         |
|---------------|----------|--------------------------------|----------|---------------------------------|
| DVP 12SA2     | X0       | START PB 1                     | Y0       | PASS LAMP                       |
|               | X1       | START PB 2                     | Y1       | BUZZER                          |
|               | X2       | EMB PB                         | Y2       | PROCESS LAMP                    |
|               | X3       | RESET PB                       | Y3       | ERROR LAMP                      |
|               | X4       | AIR PRESSURE SWITCH            | NA       | NA                              |
|               | X5       | LIGHT CURTAIN-1                | NA       | NA                              |
|               | X6       | MAIN CLY FWD RD SW             | NA       | NA                              |
|               | X7       | MAIN CLY RET RD SW             | NA       | NA                              |
| DVP 16SP11R-1 | X20      | Toggle/HOLD CLAMP 1 FWD RD     | Y20      | MAIN RAM CLY FWD SOL            |
|               | X21      | Toggle/HOLD CLAMP 1 RET RD     | Y21      | MAIN RAM CLY RET SOL            |
|               | X22      | Toggle/HOLD CLAMP 2 FWD RD     | Y22      | BUSH PRESSING CLY FWD SOL       |
|               | X23      | Toggle/HOLD CLAMP 2 RET RD     | Y23      | BUSH PRESSING CLY RET SOL       |
|               | X24      | Toggle/HOLD CLAMP 3 FWD RD     | Y24      | TOGGLE/HOLD CLAMP CLY 1 FWD SOL |
|               | X25      | Toggle/HOLD CLAMP 3 RET RD     | Y25      | TOGGLE/HOLD CLAMP CLY 1 RET SOL |
|               | X26      | Toggle/HOLD CLAMP 4 FWD RD     | Y26      | TOGGLE/HOLD CLAMP CLY 2 FWD SOL |
|               | X27      | Toggle/HOLD CLAMP 4 RET RD     | Y27      | TOGGLE/HOLD CLAMP CLY 2 RET SOL |
| DVP 16SP11R-2 | X30      | FIX SEL-1                      | Y30      | TOGGLE/HOLD CLAMP CLY 3 FWD SOL |
|               | X31      | FIX SEL-2                      | Y31      | TOGGLE/HOLD CLAMP CLY 3 RET SOL |
|               | X32      | FIX SEL-3                      | Y32      | TOGGLE/HOLD CLAMP CLY 4 FWD SOL |
|               | X33      | STEEL BUSH PRSENCE SENSOR 1    | Y33      | TOGGLE/HOLD CLAMP CLY 4 RET SOL |
|               | X34      | STEEL BUSH PRSENCE SENSOR 2    | Y34      | AUTO MODE SEL LAMP              |
|               | X35      | STEEL BUSH PRSENCE SENSOR 3    | Y35      | MANUAL MODE SEL LAMP            |
|               | X36      | STEEL BUSH PRSENCE SENSOR 4    | Y36      | JIG CHANGOVER LAMP              |
|               | X37      | STEEL BUSH PRSENCE SENSOR 5    | Y37      | CONTROL ON                      |
| DVP 16SP11R-3 | X40      | BRASS BUSH PRESENT SENSOR      | Y40      | AIR OFF SOL                     |
|               | X41      | PART PRESENT SENSOR            | Y41      | PANEL FAN                       |
|               | X42      | STEEL BUSH PRSENCE SENSOR 6    | Y42      | PANEL LIGHT                     |
|               | X43      | STEEL BUSH PRSENCE SENSOR 7    | Y43      | SPARE                           |
|               | X44      | STEEL BUSH PRSENCE SENSOR 8    | Y44      | CONTACTOR ON                    |
|               | X45      | STEEL BUSH PRSENCE SENSOR 9    | Y45      | SPARE                           |
|               | X46      | FIXTURE DOWEL PRESENT SENSOR 1 | Y46      | SPARE                           |
|               | X47      | FIXTURE DOWEL PRESENT SENSOR 2 | Y47      | SPARE                           |
|               | X50      | ROD/TOOL PRESENT SENSOR 1      |          |                                 |



## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

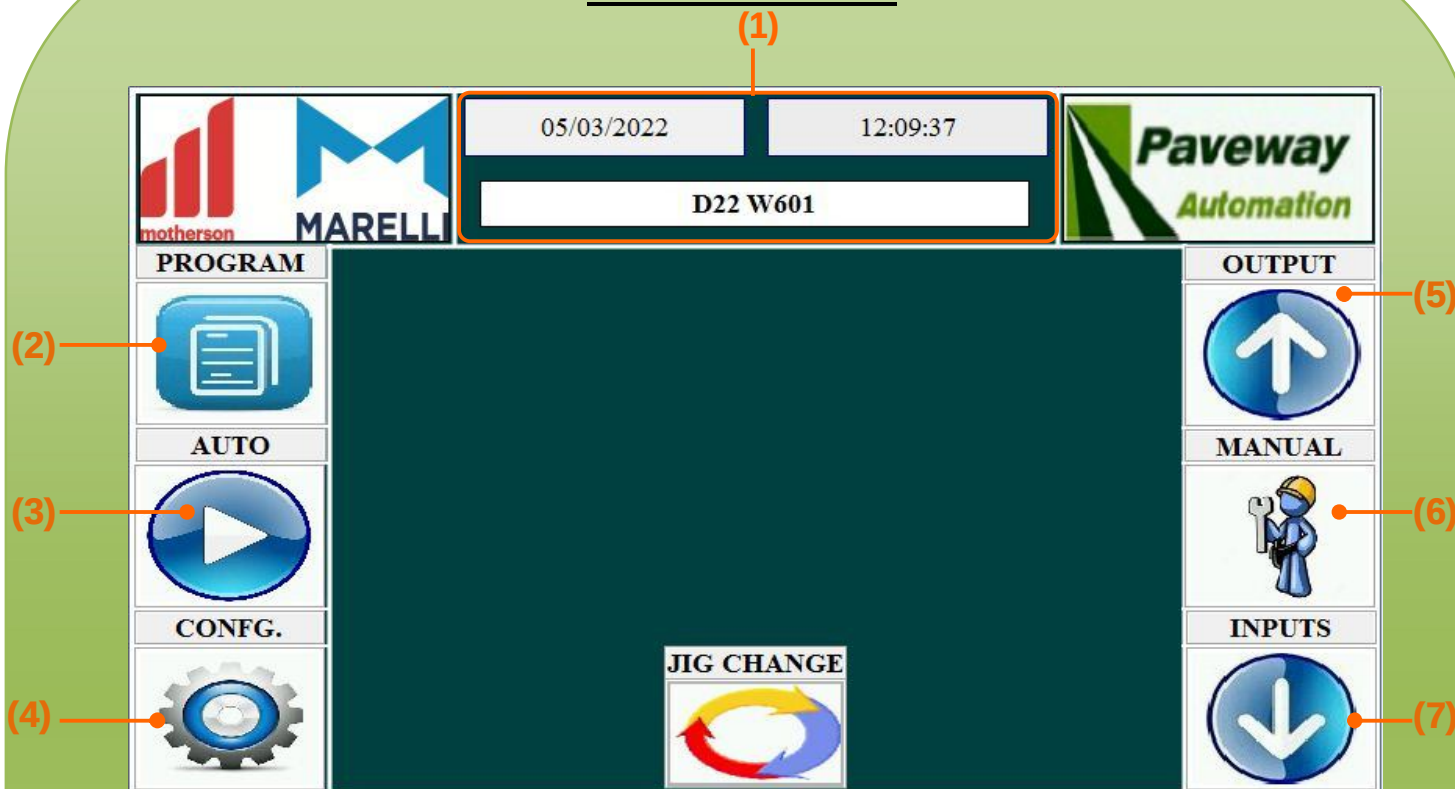
Gat No 148/150, Chakan,Pune-410501  
(Near Dwarka School ,Village- Ambethan ,Taluka- Khed)



|                |     |                                    |  |  |
|----------------|-----|------------------------------------|--|--|
| DVP<br>16SM11N | X51 | ROD/TOOL PRESENT SENSOR 2          |  |  |
|                | X52 | DOT PUNCH CLY FWD RD               |  |  |
|                | X53 | JIG REED-1                         |  |  |
|                | X54 | JIG REED-2                         |  |  |
|                | X55 | JIG REED-3                         |  |  |
|                | X56 | SPARE                              |  |  |
|                | X57 | SPARE                              |  |  |
|                | X60 | STEEL BUSH PRESS CLY 1-1 REV RD SW |  |  |
|                | X61 | STEEL BUSH PRESS CLY 1-1 FWD RD SW |  |  |
|                | X62 | DOOR LIMIT SW 1& 2                 |  |  |
|                | X63 | AUTO SELECTOR SW                   |  |  |
|                | X64 | MANUAL SELECTOR SWITCH             |  |  |
|                | X65 | JIG CHANGEOVER SWITCH              |  |  |
|                | X66 | CONTROL PANEL                      |  |  |
|                | X67 | SPARE                              |  |  |
| DVP<br>08SM11N | X70 | SPARE                              |  |  |
|                | X71 | SPARE                              |  |  |
|                | X72 | SPARE                              |  |  |
|                | X73 | SPARE                              |  |  |
|                | X74 | SPARE                              |  |  |
|                | X75 | SPARE                              |  |  |
|                | X76 | SPARE                              |  |  |
|                | X77 | SPARE                              |  |  |

## HMI SCREEN DETAILS:

### HOME SCREEN



#### 1. Information Display Area

Displays the Program name, Date & Time of currently in operation.

#### 2.Program monitoring screen button .

#### 3.Auto monitoring screen button.

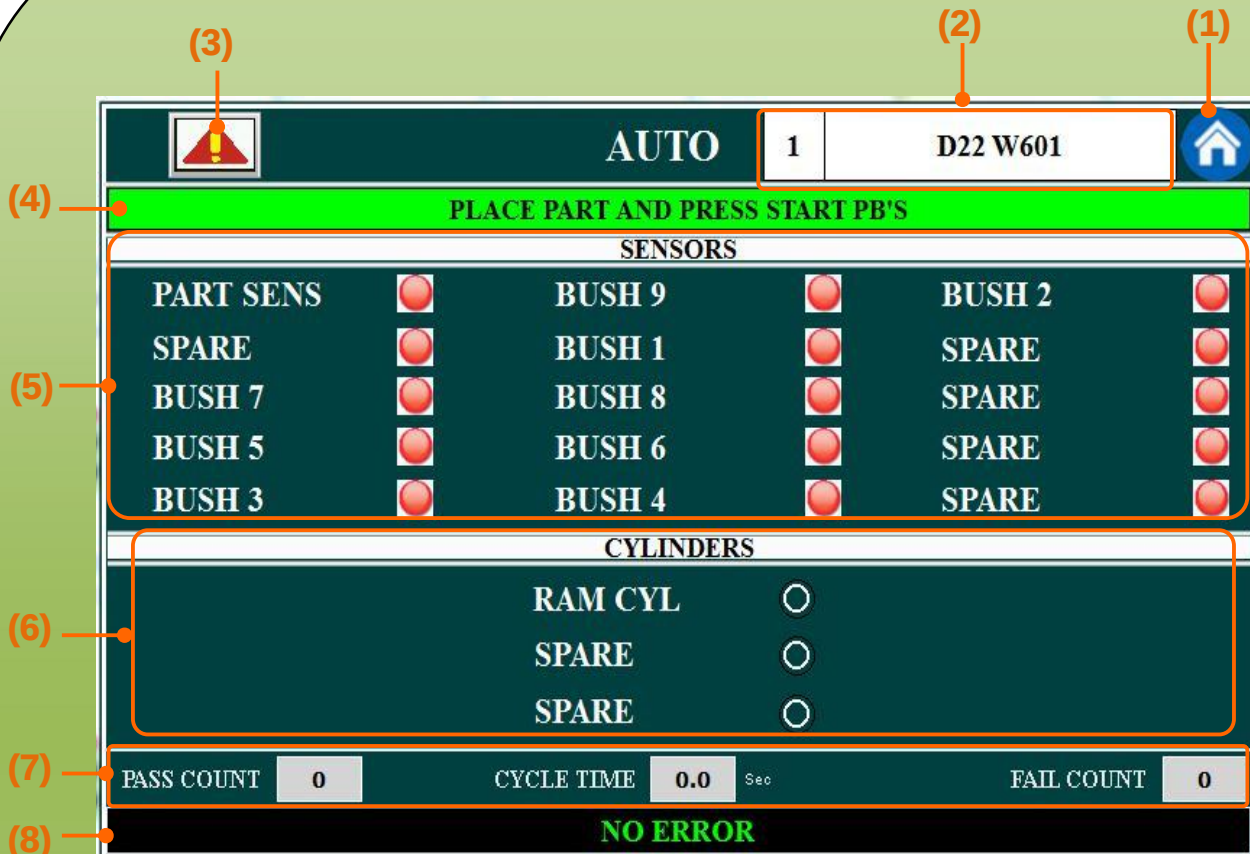
#### 4.Configuration monitoring screen button.

#### 5. Output monitoring screen button.

#### 6.Manual monitoring screen button.

#### 7. Inputs monitoring screen button..

### AUTO MODE SCREEN-LH



The screenshot shows the 'AUTO MODE SCREEN-LH' interface. At the top, there is a header bar with a warning icon (3), the word 'AUTO', a box containing '1' (2), and a box containing 'D22 W601' (2). To the right of the header is a home icon (1). Below the header is a green bar with the text 'PLACE PART AND PRESS START PB'S' (4). Underneath is a 'SENSORS' section with a table of sensor status indicators (5). Below the sensors is a 'CYLINDERS' section with a table of cylinder status indicators (6). At the bottom, there is a status bar (7) showing 'PASS COUNT 0', 'CYCLE TIME 0.0 Sec', and 'FAIL COUNT 0'. Below the status bar is a black bar with the text 'NO ERROR' (8).

| SENSORS   |                          |        |
|-----------|--------------------------|--------|
| PART SENS | <input type="checkbox"/> | BUSH 9 |
| SPARE     | <input type="checkbox"/> | BUSH 1 |
| BUSH 7    | <input type="checkbox"/> | BUSH 8 |
| BUSH 5    | <input type="checkbox"/> | BUSH 6 |
| BUSH 3    | <input type="checkbox"/> | BUSH 4 |

| CYLINDERS |                          |
|-----------|--------------------------|
| RAM CYL   | <input type="checkbox"/> |
| SPARE     | <input type="checkbox"/> |
| SPARE     | <input type="checkbox"/> |

|            |   |            |     |     |            |   |
|------------|---|------------|-----|-----|------------|---|
| PASS COUNT | 0 | CYCLE TIME | 0.0 | Sec | FAIL COUNT | 0 |
|------------|---|------------|-----|-----|------------|---|

NO ERROR

- 1.Home screen navigation button.
2. Program name & Number.
3. Navigation button.
4. Operation status display.
5. Sensor status display.
6. Cylinders status display.
- 7.Pass count, cycle time, fail count operation status display.
8. Error message display.

### CONFIGURATION SCREEN

CONFIGURATION

1

D22 W601



(1)

SENSORS

(2)  PART SENS 

 SPARE 

 BUSH 7 

 BUSH 5 

 BUSH 3 

 BUSH 9 

 BUSH 1 

 BUSH 8 

 BUSH 6 

 BUSH 4 

 BUSH 2 

 SPARE 

 SPARE 

 SPARE 

 SPARE 

| CYLINDER             | FWD REED S/W                                                                        | RET REED S/W                                                                        | TIME (s) | HOLD TIME (s) | CYLINDER  | REED S/W                                                                              | TIME (s)                                                                              | HOLD TIME (s) |
|----------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------|---------------|-----------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|
| RAM                  |  |  | 12.0     | 6.0           | SPARE     |  |  | 0.0           |
|                      |  |  |          |               | FIX CLAMP |  |                                                                                       | 2.0           |
| RAM AIR CUT OFF TIME |                                                                                     |                                                                                     | 3.0      |               |           |                                                                                       |                                                                                       |               |
| SPARE                |  |  | 0.0      | 0.0           |           |                                                                                       |                                                                                       |               |

(3) (4)

SYSTEM SETUP

DATE & TIME

PASSWORD

LIGHT CURTAIN 

FIXTURE NO.: 3

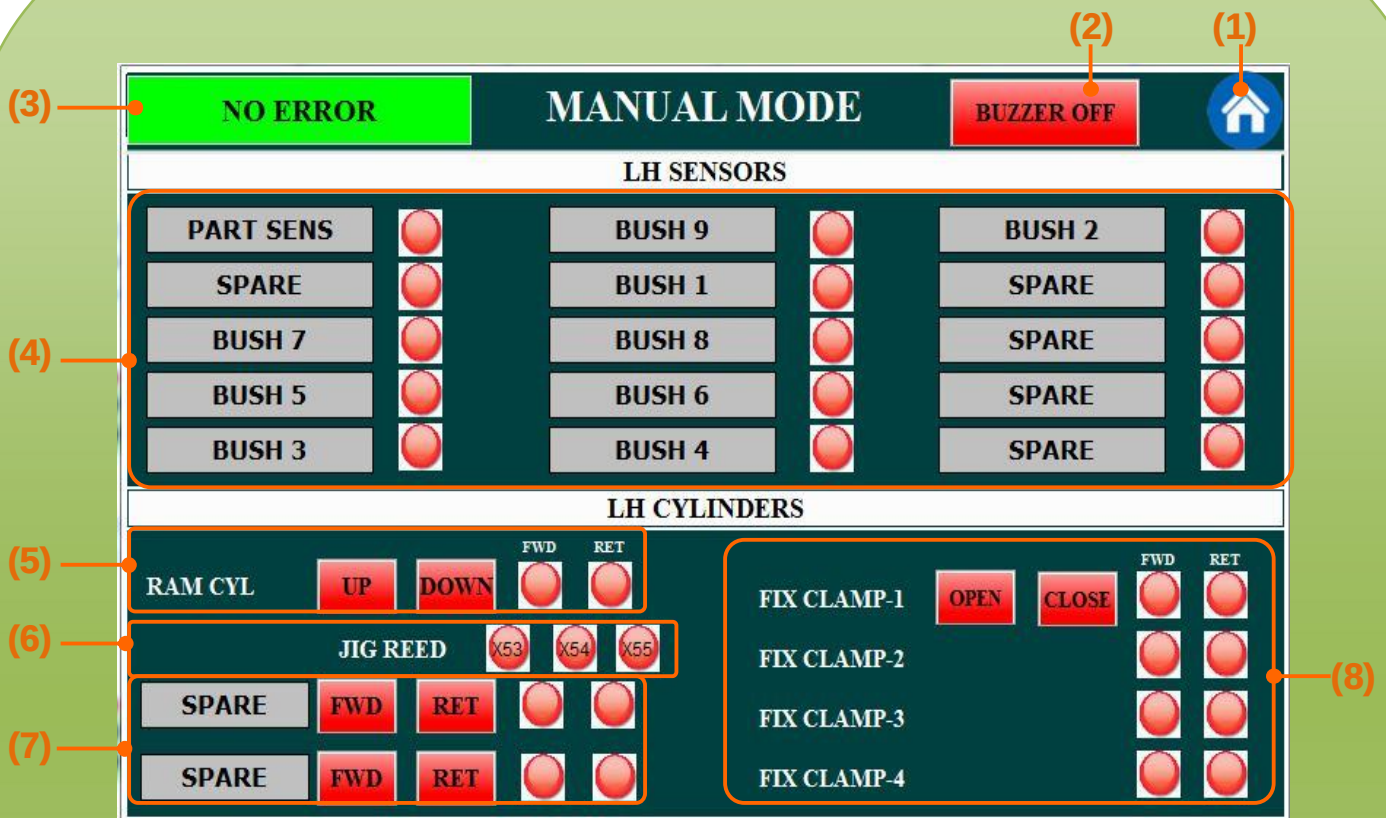
(6)

(7)

1. Home screen navigation button, Program name & Number display.
2. Sensors select/unselect setting option.
3. Ram & Fix clamp Cylinder reed switch select/unselect option and setting time in sec.
4. Ram cylinder cut-off time display.
5. System setup
 

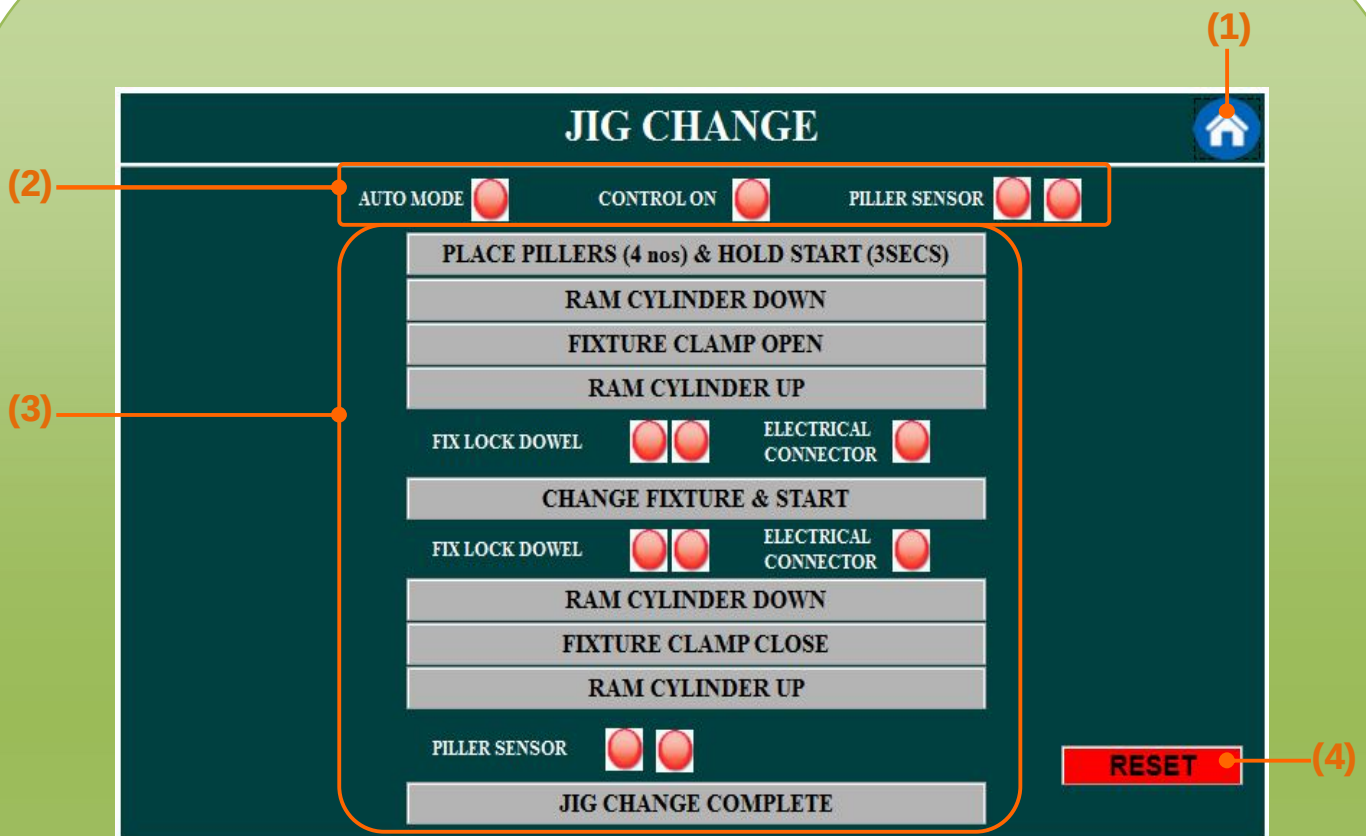
Date & time, Password setting button.
6. Light curtain status display.
7. Fixture no display.

### MANUAL SCREEN



1. Home screen navigation button.
2. Buzzer off button.
3. Error status message display.
4. Part present sensor & 1 to 9 Bush manually operated mode display.
5. Ram cylinder manually operation status display.
6. Jig reed input display.
7. Spare cylinders.
8. Fix clamp 1 to 4 cylinders manually operation status display.

### JIG CHANGE



1. Home screen navigation button.
2. Auto mode, Control on, Piller sensor status display..
3. Jig change process display.
4. Reset button.

### INPUT STATUS SCREEN-1

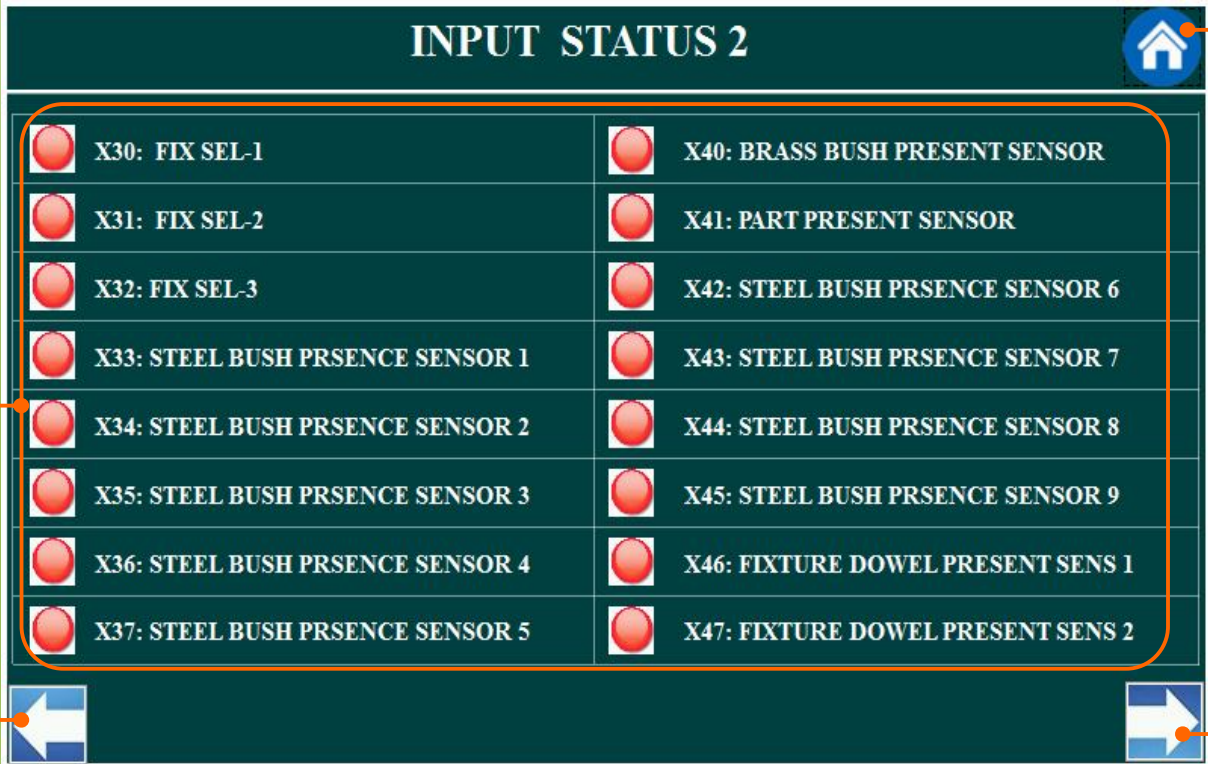
| INPUT STATUS 1                                                                                             |                                                                                                                     | (1) |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----|
|  X0: START PB 1           |  X20: Toggle/HOLD CLAMP 1 FWD RD   | (2) |
|  X1: START PB 2           |  X21: Toggle/HOLD CLAMP 1 RET RD   |     |
|  X2: EMB PB               |  X22: Toggle/HOLD CLAMP 2 FWD RD   |     |
|  X3: RESET                |  X23: Toggle/HOLD CLAMP 2 RET RD   |     |
|  X4: AIR PRESSURE SWITCH |  X24: Toggle/HOLD CLAMP 3 FWD RD  |     |
|  X5: LIGHT CURTAIN 1    |  X25: Toggle/HOLD CLAMP 3 RET RD |     |
|  X6: MAIN CLY FWD RD SW |  X26: Toggle/HOLD CLAMP 4 FWD RD |     |
|  X7: MAIN CLY RET RD SW |  X27: Toggle/HOLD CLAMP 4 RET RD |     |
|                                                                                                            |                                                                                                                     | (3) |

1. Home screen navigation button.

2. PLC input status.

3. Next page navigation button.

### INPUT STATUS SCREEN -2



| INPUT STATUS 2                                                                                                       |                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|  X30: FIX SEL-1                     |  X40: BRASS BUSH PRESENT SENSOR      |
|  X31: FIX SEL-2                     |  X41: PART PRESENT SENSOR            |
|  X32: FIX SEL-3                    |  X42: STEEL BUSH PRSENCE SENSOR 6   |
|  X33: STEEL BUSH PRSENCE SENSOR 1 |  X43: STEEL BUSH PRSENCE SENSOR 7  |
|  X34: STEEL BUSH PRSENCE SENSOR 2 |  X44: STEEL BUSH PRSENCE SENSOR 8  |
|  X35: STEEL BUSH PRSENCE SENSOR 3 |  X45: STEEL BUSH PRSENCE SENSOR 9  |
|  X36: STEEL BUSH PRSENCE SENSOR 4 |  X46: FIXTURE DOWEL PRESENT SENS 1 |
|  X37: STEEL BUSH PRSENCE SENSOR 5 |  X47: FIXTURE DOWEL PRESENT SENS 2 |

1. Home screen navigation button.
2. PLC input status.
3. Previous page navigation button.
4. Next page navigation button.

### INPUT STATUS SCREEN -3

| INPUT STATUS 3                                                                                                   |                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|  X50: ROD/TOOL PRESENT SENSOR 1 |  X60: STEEL BUSH PRESS CLY 1-1 REV RD SW |
|  X51: ROD/TOOL PRESENT SENSOR 2 |  X61: STEEL BUSH PRESS CLY 1-1 FWD RD SW |
|  X52: DOT PUNCH CLY FWD RD      |  X62: DOOR LIMIT SW 1& 2                 |
|  X53: JIG REED-1                |  X63: AUTO SELECTOR SW                   |
|  X54: JIG REED-2               |  X64: MANUAL SELECTOR SWITCH            |
|  X55: JIG REED-3              |  X65: JIG CHANGEOVER SWITCH            |
|  X56: SPARE                   |  X66: CONTROL PANEL                    |
|  X57: SPARE                   |  X67: SPARE                            |

1. Home screen navigation button.

2. PLC input status.

3. Previous page navigation button.

### OUTPUT STATUS SCREEN -1

#### OUTPUT STATUS 1



(1)

|                                                                                                                   |                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|  Y0: PASS LAMP                   |  Y20: MAIN RAM CLY FWD SOL              |
|  Y1: BUZZER                      |  Y21: MAIN RAM CLY RET SOL              |
|  Y2: PROCESS LAMP                |  Y22: BUSH PRESSING CLY FWD SOL         |
|  Y3: ERROR                      |  Y23: BUSH PRESSING CLY FWD SOL        |
|  %QX0.4: AUTO MODE INDICATOR   |  Y24: TOGGLE/HOLD CLAMP CLY 1 FWD SOL |
|  %QX0.5: MANUAL MODE INDICATOR |  Y25: TOGGLE/HOLD CLAMP CLY 1 RET SOL |
|  %QX0.6: SPARE                 |  Y26: TOGGLE/HOLD CLAMP CLY 2 FWD SOL |
|  %QX0.7: SPARE                 |  Y27: TOGGLE/HOLD CLAMP CLY 2 RET SOL |



(3)

1. Home screen navigation button.
2. Plc output status.
3. Next page navigation button.

### OUTPUT STATUS SCREEN -2

| OUTPUT STATUS 2                                                                                                        |                                                                                                    | (1)                                                                                       |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  Y30: TOGGLE/HOLD CLAMP CLY 3 FWD SOL |  Y40: AIR OFF SOL |                                                                                           |
|  Y31: TOGGLE/HOLD CLAMP CLY 3 RET SOL |  Y41: PANEL FAN   |                                                                                           |
|  Y32: TOGGLE/HOLD CLAMP CLY 4 FWD SOL |  Y42: PANEL LIGHT |                                                                                           |
|  Y33: TOGGLE/HOLD CLAMP CLY 4 RET SOL |  Y43: SPARE       |                                                                                           |
|  Y34: AUTO MODE SEL LAMP            |  Y44: SPARE     |                                                                                           |
|  Y35: MANUAL MODE SEL LAMP          |  Y45: SPARE     |                                                                                           |
|  Y36: JIG CHANGOVER LAMP            |  Y46: SPARE     |                                                                                           |
|  Y37: CONTROL ON                    |  Y47: SPARE     |                                                                                           |
| (3)                                 |                                                                                                    |  (4) |

1. Home screen navigation button.
2. Plc output status.
3. Previous page navigation button.
4. Next page navigation button.

### OUTPUT STATUS SCREEN -3

| OUTPUT STATUS 3                                                                             |  |
|---------------------------------------------------------------------------------------------|--|
|  Y50: NA   |  |
|  Y51: NA   |  |
|  Y52: NA   |  |
|  Y53: NA |  |
|  Y54: NA |  |
|  Y55: NA |  |
|  Y56: NA |  |
|  Y57: NA |  |

(1)

(2)

(3)

1. Home screen navigation button.
2. Plc output status.
3. Previous page navigation button.



## D22 AIM COMPRESSION LIMITER PRESSING MACHINE.

Gat No 148/150, Chakan,Pune-410501  
(Near Dwarka School ,Village- Ambethan ,Taluka-Khed)



### **DO'S & DONT'S:**

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.

- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting into electrical connector.



## **D22 AIM COMPRESSION LIMITER PRESSING MACHINE.**

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